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ORIGINAL PAPER

Manipulation of moves in sequential contests

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Abstract
In two-player Tullock contests with endogenous timing of moves, the weak player moves early and the strong player follows. This order of moves is the third-best outcome for a contest organiser as it leads to a contest with lower aggregate effort compared to a contest where the players move in reverse order (first best) or simultaneously (second best). We propose that if the contest organiser—who does not know *ex ante* which player is the strong and which is the weak one—offers a lower price (effort cost) to the player(s) who choose(s) to exert effort early, she can achieve a greater payoff by affecting the contestants' sequence of moves. We show that there exists no price that generates the first-best outcome. However, there is a price (or a range of prices) that induces both players to move early and receive it, leading to the second-best outcome. We also discuss the case where both players move early but only one receives the low price, e.g., lobbying or Instagram "giveaway" contests.

1 Introduction

There are many situations in which a number of contestants compete with each other exerting costly efforts in order to win a prize. These efforts include, among others, monetary expenditures or time spending, while the prize awarded may include monopoly rents, sports awards or fame.[1] While most of the existing research has focused on simultaneous contests, there are cases in which the contestants' efforts take place sequentially rather than simultaneously and the sequence of their moves is out of the contest organiser's control. For instance, during election campaigning, an interest group might choose the level of monetary contribution to offer to its favourite candidate before the opposing group does so. Similarly, in a "giveaway"

[1] For surveys of the literature, see Nitzan (1994), Corchón (2007), Konrad (2009) and Corchón and Serena (2017).

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contest on Instagram, when a user enters the contest and chooses how much effort to exert, she already knows the other participants' levels of effort.[2]

In this study, we consider a Tullock contest in which two individuals compete with each other to win a prize. The individuals, who have different valuations of the prize, have the opportunity to exert effort either early or late, i.e., the contestant who moves late observes the early mover's effort before she chooses how much effort to put (Dixit 1987). After the levels of effort have been observed, the two individuals compete with each other in a Tullock contest (Tullock 1980). The winner of the contest receives the prize.

It is a well-known result in the contest theory literature that in the equilibrium of a sequential Tullock contest where the players can endogenously decide the timing of their moves, they agree that the player who values the prize lower (weak player) exerts effort early and the one who values the prize higher (strong player) follows (Baik and Shogren 1992; Leininger 1993). This way, both players benefit because the weak player—by moving early—chooses a moderate level of effort and then, the strong player replies choosing a level of effort greater but still lower than the one in the case where she moves early. While the situation described above is mutually beneficial for the two players as it leads to a lower level of aggregate expenditure, this is not the case for an effort-maximising contest organiser.[3] Compared to the sequence of moves chosen by the players, a contest organiser prefers either a situation where the players move in an inverse way, i.e., the strong player moves early and the weak one follows (first best) or at least a situation where the players move simultaneously (second best) (Linster 1993). For instance, in the case of informational lobbying, a politician, i.e., the contest organiser in our model, might prefer the strongest interest group to make a lobbying effort[4] early since this would overall lead to a greater aggregate level of information.[5] Likewise, the organiser of an Instagram "giveaway" contest prefers a situation where the users who value the prize higher or have many followers (strong players) participate early in the contest because it is more likely that they will communicate the contest to more users than the weaker players will.

Following the above discussion, the main purpose of this study is to investigate how a contest organiser can increase her expected payoff by being an active part of the contest and manipulating the (agreed) order of the players' moves. We consider, therefore, a Tullock contest with two players who have different prize valuations. Then, we make the assumption that the contest organiser, who values early effort

[2] For more information on "giveaway" contests, see https://bit.ly/3jBeerB and https://bit.ly/3A93Tbj.

[3] Following the majority of the relevant literature, we consider a contest organiser that seeks to maximise total effort. For example, if we assume that interest groups make effort in order to gather information beneficial to the politician, then, the latter might want to maximise the sum of efforts and, as a consequence, receive as much information as possible.

[4] In such a situation, lobbying effort by an interest group can be viewed as resources spent for research on a relevant policy issue or time spent providing information to a politician about this issue.

[5] See Cotton and Déllis (2016) and Cotton and Li (2018) for some relevant studies on informational lobbying.

higher than late effort, offers a lower price per unit of effort to the player that commits to participate early in the contest. This lower price per unit of effort offered by the contest organiser to the early mover can be viewed, for instance in the lobbying example, as a lower "access" price per hour spent in the politician's office for the group that makes a lobbying effort early. Alternatively, it can be perceived as a greater weight a politician puts on the lobbying effort of this particular group when deciding the winner of the final prize, e.g., the assignment of a project, trade protection or the adoption of the group's favourite policy.[6] In the Instagram "giveaway" example, the lower price can be viewed as an advantage offered by the contest organiser to the early movers by announcing that the ones entering the contest within a specific period of time after the announcement of it will receive extra entries in the contest and, therefore, will have a greater probability of winning the prize. In general, the contest organiser's objective is to set the level of price that not only yields the maximum possible payoff for her but also induces the players to agree to the sequence of moves that generates this particular level of payoff.

Beginning with the benchmark model, we assume that if both players decide to move together early, they both receive the low price. Then, we study the players' optimal behaviour and determine the optimal level of the price set by the contest organiser. Our findings suggest that there exists no pure-strategy Nash equilibrium in which the strong player exerts effort early and the weak player follows. However, under the optimal level of price for the early mover, the contest organiser can achieve a greater payoff than in the case with no intervention by making both players exert effort early. Actually, as the exact level of price does not affect her expected payoff, the contest organiser is indifferent among a range of prices as long as they induce a simultaneous order of moves by the players. This simultaneous order of moves by the two players occurs because the price set by the contest organiser for the early mover induces not only the weak but also the strong player to move early—and, therefore, pay the low price—instead of giving up the low price by waiting for the former to be the early mover.

Next, we modify the benchmark model by assuming that if the two players decide to move early, only one of them pays the low price with some probability. For instance, in the lobbying example, while we assume that the politician definitely values the lobbying effort that takes place early higher than the later one when they take place simultaneously, she might want to favour the group that is closer to her political preferences by offering a low price to it with a greater probability. Furthermore, in the example of the "giveaway" contest, even when a user participates in the contest by writing a comment on a post without having seen the other users' replies, the comments of the post will appear below each other. Thus, the contest organiser may offer an advantage only to the user who has been faster at typing and her comment appears first on the list of comments. In contrast to the benchmark model, in

[6] The concept of price-setter candidates is similar to Austen-Smith (1998) where an incumbent politician commits to the contribution level she requires from groups in order to consider them as potential information providers.

this case, the level of the price affects the contest organiser's payoff even when the players move simultaneously. Thus, we can determine the exact price set by the contest organiser which now depends also on the players' probabilities of receiving it when they both move early.

1.1 Link to the literature

The paper mainly contributes to the literature on contests and particularly endogenous timing in sequential contests. In a seminal study, Dixit (1987) considers a two-player contest where the players can strategically pre-commit to their levels of effort. He finds that if the favourite player has the opportunity to move early, she commits to a level of effort greater than the Nash equilibrium.[7] In a later study, Yildirim (2005) considers a case where contestants can exert effort multiple times and finds that the outcome with the underdog's leadership can never be an equilibrium.

Introducing endogenous timing, Baik and Shogren (1992) and Leininger (1993) allow the players to choose their order of moves before they exert any effort. Interestingly, they show that the favourite player prefers to wait for her opponent to move early and the underdog finds this order of moves advantageous for her too.[8] Fu (2006) extends the above studies introducing asymmetric information across players and finds that, in equilibrium, the informed player moves second. In a later study, Hoffmann and Rota-Graziosi (2012) also consider the case with an endogenous prize and show that players may decide to play simultaneously in the sub-game perfect equilibrium. Finally, Morgan (2003) shows that—under specific assumptions (Serena 2017)—even if the two players are *ex ante* identical, they choose to exert efforts sequentially.

Some other studies related to ours are the ones on contests with reimbursements (Baik and Shogren 1994; Matros and Armanios 2009; Matros 2012). The former considers an environmental conflict between a firm and a group of citizens where the legal expenses of the latter are reimbursed if it wins the contest. The other two investigate a n-player symmetric Tullock contest with winner/loser effort reimbursements and an asymmetric winner-reimbursed contest, respectively. Moreover, Minchuk (2018) studies the effect of a winner's reimbursement on all-pay auctions while Liu and Liu (2019) consider a winner's reimbursement in an all-pay auction with risk-averse contestants. In contrast to the above-mentioned studies which assume an effort reimbursement to the winner and/or loser of a one-shot contest, we consider a situation where an advantage (lower price) is set *endogenously* and given to the player that moves early in a sequential contest with endogenous timing of moves.[9]

[7] See also Baye and Shin (1999) for an elaboration on the second-order conditions of the symmetric case in Dixit (1987).

[8] See also Glazer and Hassin (2000) and Hinnosaar (2023) who study sequential Tullock contests with more than two periods.

[9] Baik and Shogren (1994) also explore endogenous timing of moves assuming that the favourite contestant is the one that *exogenously* receives a greater reimbursement in case of winning the contest, i.e., the group of citizens. They find that the group of citizens and the firm move simultaneously if the reimbursement is low while the firm moves first exerting zero effort otherwise.

Another study relative to this one is Cohen and Sela (2005) in which the designer of a Tullock contest manipulates the players' probabilities of winning by reimbursing the winner's cost of effort. The result of the introduction of such reimbursement is that the weak contestant has a greater probability of winning the contest than the strong one. Different to the above study, in this paper, the players have the opportunity to move sequentially. Moreover, the low price offered by the contest organiser aims at manipulating the order of moves decided by the players and, consequently, the contest organiser's expected payoff.

Finally, a study close to this one is Protopappas (2022) which considers an election-lobbying game with two candidates who, before the election takes place, announce the optimal "access" price the interest groups have to pay in order to have the opportunity to simultaneously lobby the eventual winner of the election. In contrast to Protopappas (2022) where in the contest between the two groups, the lobbying efforts take place simultaneously, in this paper, we consider a setting where the contestants may also choose to move sequentially.

The remainder of the paper is structured as follows. In Sect. 2, we present an overview of the model. In Sect. 3, we analyse the model and provide the results of the analysis. In Sect. 4, we study the extension of the benchmark model discussed above. Lastly, Sect. 5 concludes the paper. All proofs are included in Appendix 1.

2 Outline of the model

Consider a Tullock contest consisting of a contest organiser who offers a prize and two players who compete with each other in order to win this prize. Player i's valuation of the prize is common knowledge among the agents and equal to $v_i > 0$ where $i \in \{1, 2\}$. Without loss of generality, we assume that player 1 values the prize higher than player 2, i.e., $v_1 > v_2$. Player 1 and player 2, having as an objective the maximisation of their expected payoffs, exert costly efforts, $x_1 \geq 0$ and $x_2 \geq 0$, respectively, in order to increase their probability of winning the prize. Making use of the popular lottery contest success function proposed by Tullock (1980) and axiomatised by Skaperdas (1996) and Clark and Riis (1998), we assume that each player's probability of winning the prize equals the ratio of this player's effort to the sum of efforts. Thus, when player i makes effort $x_i,\ i = 1, 2$, her probability of winning is[10]

$$p_i\left(x_i, x_j\right) = \begin{cases} \dfrac{x_i}{x_i + x_j}, & \text{if } \max\{x_i, x_j\} > 0, \quad i \neq j \\ \dfrac{1}{2}, & \text{otherwise.} \end{cases} \tag{1}$$

Before the contest begins, the contest organiser announces that the player who commits to exert effort early is going to pay a price $c \in [0, 1]$ per unit of effort. Following

[10] This contest success function is a special case of $p_i\left(x_i, x_j\right) = \frac{x_i^r}{x_i^r + x_j^r}$ with $r = 1$. For $r \neq 1$, because of the sequential setting, the analysis becomes much more complicated and, inevitably, intractable.

the discussion in the Introduction regarding a contest organiser's preference for early effort, we assume that the contest organiser values the effort exerted early higher than the one exerted late, i.e., she discounts late effort by a factor $\delta \in [0, 1]$. Her expected payoff is then

$$\pi_{o,E}\left(x_{1,E}, x_{2,E}, c\right) = cx_{1,E} + cx_{2,E} \tag{2}$$

in the case where both players move early and pay price c;

$$\pi_{o,L}\left(x_{1,L}, x_{2,L}\right) = \delta\left(x_{1,L} + x_{2,L}\right) \tag{3}$$

in the case where both players move late and pay a price equal to 1;

$$\pi_{o,ij}\left(x_{i,ij}, x_{j,ij}, c\right) = cx_{i,ij} + \delta x_{j,ij} \tag{4}$$

in the case of a sequential contest where $x_{i,ij}$ is the effort exerted by player i when she moves early and player j follows. The contest organiser's objective is to set the level of price that yields the maximum expected payoff for her.[11]

Let us now further discuss our pricing mechanism. A contest organiser, instead of setting *ex ante* a lower price for the early mover, could alternatively set a different price for each player. However, this case is not always feasible because individual communication between the contest organiser and a player might be costly. For instance, it is less costly—and more common in real life—for an Instagram "giveaway" contest organiser to announce a price for the early mover—whoever this is—rather than assigning a specific price to each participant. Another situation where such a pricing mechanism might emerge is when a contest organiser knows that there are players with different levels of "strength" but cannot distinguish to which player each level corresponds.

The timing of the game is the following. At the first stage, the contest organiser announces the level of price for the player who is going to exert effort early. At the second stage, the two players, having observed the level of the price announced by the contest organiser, decide on the timing of their moves, namely, whether to exert effort early or late. At the final stage and given the order of the moves decided, the sequential or simultaneous Tullock contest takes place and the winner receives the prize. We use backward induction to determine the sub-game perfect Nash equilibrium in pure strategies.

[11] As we will discover later, the contest organiser's maximum expected payoff occurs when both players move early, i.e., maximising (2). Therefore, the existence of the discount factor δ eventually does not affect the main result of the paper, i.e., the contest organiser's optimal choice of price.

3 Analysis and results

3.1 Third stage (Contest)

At the beginning of the third stage, the level of price for the early mover has already been set and the players have decided on the sequence of their moves. Following Leininger (1993), we assume that when the two players' timing decisions coincide, they play a simultaneous game. In the rest of the section, we analyse the four possible cases of the third-stage game.

3.1.1 Both players move early (*E*)

This is the most common case in the literature where each player makes effort without knowing the other player's level of effort. We assume that since both players move early, they both pay the price c announced by the contest organiser. Player i's maximisation problem is, then,

$$\max_{x_i} \left\{ p_i\left(x_i, x_j\right) v_i - c x_i \right\}. \tag{5}$$

where $p_i\left(x_i, x_j\right)$ is given by (1). Maximising the above expression with respect to x_i, we obtain that in the equilibrium of the simultaneous game where both players move early, player i's effort is

$$x^*_{i,E} = \frac{v_i^2 v_j}{c(v_i + v_j)^2}, \qquad i,j = 1,2, \quad i \neq j \tag{6}$$

and her expected payoff is

$$\pi^*_{i,E} = \frac{v_i^3}{\left(v_i + v_j\right)^2}, \qquad i,j = 1,2, \quad i \neq j.$$

3.1.2 Both players move late (*L*)

In this case, similarly to the previous one, the two players move simultaneously. However, because they move late, they both pay a price equal to 1. Player i's maximisation problem is, then,

$$\max_{x_i} \left\{ p_i\left(x_i, x_j\right) v_i - x_i \right\}. \tag{7}$$

The above problem is a typical two-player Tullock contest with asymmetric valuations (see Nti (1999)). In equilibrium, we have that

$$x^*_{i,L} = \frac{v_i^2 v_j}{(v_i + v_j)^2}, \qquad i,j = 1,2, \quad i \neq j$$

and

$$\pi^*_{i,L} = \frac{v_i^3}{\left(v_i + v_j\right)^2}, \qquad i,j = 1,2, \quad i \neq j.$$

3.1.3 Player 1 moves early, player 2 moves late (*EL*)

In the first sequential case, let us assume that the two players have decided that player 1 makes effort early and player 2 follows. Assume, also, that the contest organiser has announced that the early mover will pay a price $c \in [0, 1]$ per unit of effort. Then, player 1 exerts a publicly observable effort, $x_{1,12}$, and player 2, knowing her opponent's level of effort, makes her move exerting effort $x_{2,12}$. Player 2's maximisation problem is

$$\max_{x_{2,12} \geq 0} \left\{ \frac{x_{2,12}}{x_{1,12} + x_{2,12}} v_2 - x_{2,12} \right\}. \tag{8}$$

Player 1 makes effort anticipating her opponent's reply. Thus, her optimisation problem is

$$\max_{x_{1,12} \geq 0} \left\{ \frac{x_{1,12}}{x_{1,12} + x_{2,12}\left(x_{1,12}\right)} v_1 - c x_{1,12} \right\}, \tag{9}$$

where c is the price player 1 is charged because she moves early and $x_{2,12}\left(x_{1,12}\right)$ is player 2's reaction function. Differentiating the above expression with respect to $x_{1,12}$, setting the derivative equal to zero and solving for $x_{1,12}$ gives us player 1's equilibrium effort, $x^*_{1,12}$. Then, substituting player 1's equilibrium effort into player 2's reaction function, we can also find player 2's equilibrium effort, $x^*_{2,12}$. Finally, substituting the two players' equilibrium efforts into the two players' payoff functions, we obtain their equilibrium expected payoffs. The results are summarised in the following lemma.

Lemma 1 (Leininger 1993) *When player* 1 *moves early and player* 2 *moves late, their equilibrium efforts are*

$$\left(x^*_{1,12}, x^*_{2,12}\right) = \begin{cases} \left(\frac{v_1^2}{4c^2 v_2}, \frac{v_1\left(2cv_2 - v_1\right)}{4c^2 v_2}\right), & \text{if } v_1 < 2v_2 \text{ and } c \geq \frac{v_1}{2v_2} \\ \left(v_2, 0\right), & \text{if } \left(v_1 < 2v_2 \text{ and } c \leq \frac{v_1}{2v_2}\right) \text{ or } v_1 \geq 2v_2 \end{cases}$$

and their equilibrium payoffs are

$$\left(\pi^*_{1,12},\, \pi^*_{2,12}\right) = \begin{cases} \left(\frac{v_1^2}{4cv_2},\, \frac{(v_1-2cv_2)^2}{4c^2v_2}\right), & \text{if } v_1 < 2v_2 \text{ and } c \geq \frac{v_1}{2v_2} \\ \left(v_1 - cv_2,\, 0\right), & \text{if } \left(v_1 < 2v_2 \text{ and } c \leq \frac{v_1}{2v_2}\right) \text{ or } v_1 \geq 2v_2. \end{cases}$$

The above result is in line with Leininger (1993). We observe that when player 1 is not sufficiently strong and the price announced by the contest organiser is sufficiently high, player 2 finds it optimal to participate in the contest. However, if player 1 is sufficiently strong or the price—which makes player 1 even stronger—is sufficiently low, player 2 chooses not to exert any effort ending up with a payoff equal to zero.

3.1.4 Player 2 moves early, player 1 moves late (*LE*)

In the second possible sequential case, the two players have agreed on a sequence of moves where player 2 makes effort early and player 1 responds to this effort. Following a similar process to the one in the previous case, we obtain the results stated in the following lemma.

Lemma 2 (Leininger 1993) *When player* 2 *moves early and player* 1 *moves late, their equilibrium efforts are*

$$\left(x^*_{2,21},\, x^*_{1,21}\right) = \begin{cases} \left(\frac{v_2^2}{4c^2v_1},\, \frac{v_2(2cv_1-v_2)}{4c^2v_1}\right), & \text{if } c \geq \frac{v_2}{2v_1} \\ \left(v_1,\, 0\right), & \text{if } c \leq \frac{v_2}{2v_1} \end{cases} \tag{10}$$

and their corresponding expected payoffs are

$$\left(\pi^*_{2,21}, \pi^*_{1,21}\right) = \begin{cases} \left(\frac{v_2^2}{4cv_1},\, \frac{(v_2-2cv_1)^2}{4c^2v_1}\right), & \text{if } c \geq \frac{v_2}{2v_1} \\ \left(v_2 - cv_1,\, 0\right), & \text{if } c \leq \frac{v_2}{2v_1}. \end{cases}$$

Different to the previous case, in this scenario, it is possible that player 1—the *ex ante* strong player—stays inactive. This may occur when the price offered to the early mover—player 2 in this case—is low enough to make her sufficiently stronger than player 1 and discourage the latter from participating in the contest. We observe that although player 1 values the prize higher than player 2, a sufficiently low price for the latter can make her the absolute favourite leaving the former out of the game.

3.2 Second stage (Endogenous timing)

At this stage, the two players decide on the timing of their moves. Table 1 illustrates the normal form of the extended game after replacing the sub-games with the unique

Table 1 Player 1 and player 2's expected payoffs

Player 1	Player 2	
	E	L
E	$\pi^*_{1,\text{sim}}$, $\pi^*_{2,\text{sim}}$	$\pi^*_{1,12}$, $\pi^*_{2,12}$
L	$\pi^*_{1,21}$, $\pi^*_{2,21}$	$\pi^*_{1,\text{sim}}$, $\pi^*_{2,\text{sim}}$

solutions. Note that Table 1 is not unique but there are multiple tables depending on the conditions on v_i and c and the corresponding levels of $\pi^*_{i,ij}$ and $\pi^*_{j,ij}$. For convenience, we assume that if the players are indifferent between moving early or late, they choose to do the former. Equilibrium analysis yields the following result.

Lemma 3 *In the equilibrium of the timing game, if* $c \leq \bar{c} = \frac{v_1+v_2}{2v_1}$, *both players move early* (E, E). *If* $c > \bar{c} = \frac{v_1+v_2}{2v_1}$, *player* 2 *moves early and player* 1 *moves late* (L, E).

We observe in Lemma 3 that if the price announced by the contest organiser is sufficiently high, the two players do not deviate from the optimal order of moves with the absence of a lower price offered to the early mover, i.e., the weak player moves early and the strong player follows. However, if the price the early mover gets is sufficiently low, not only the weak but also the strong player has the incentive to move early in order to gain the advantage. Furthermore, it is not difficult to observe that $\frac{\partial \bar{c}}{\partial v_1} < 0$ and $\frac{\partial \bar{c}}{\partial v_2} > 0$. This indicates that the stronger player 1 becomes, the lower the price threshold below which she decides not to wait for player 2 to make an early effort but, instead, she moves early together with her opponent. To put it differently, as player 1 becomes sufficiently strong, she prefers to wait for player 2 to move early even though the latter is the one who will pay the low price. On the other hand, as player 2 becomes stronger, the greater the price that is sufficient to induce player 1 to be willing to move early and, by doing so, prevent player 2 from paying the low price which would make her even stronger.

3.3 First stage (Pricing)

At the beginning of the game, the contest organiser chooses the level of price to set for the player who commits to exert effort early. We have already demonstrated that the level of this price can affect the players' payoffs by altering their order of moves in the contest. In the following, we can rule out the case where player 1 moves early and player 2 follows as well as the case where both players move late as these outcomes never arise in the equilibrium of the timing game for any price set by the contest organiser.[12] Hence, we are only left with two cases: the case where the two

[12] From (3), we have that $\pi^*_{o,L}\left(x^*_{1,L}, x^*_{2,L}\right) = \frac{\delta v_1 v_2}{v_1+v_2} \leq \frac{v_1 v_2}{v_1+v_2} = \pi^*_{o,E}\left(x^*_{1,E}, x^*_{2,E}\right)$, for $\delta \in [0, 1]$. Thus, even if it was optimal for both players to move late, the contest organiser would prefer an outcome where they both move early.

players move early as well as the case where player 2 and player 1 exert efforts early and late, respectively.

In the case where both players move early, the contest organiser's expected payoff is given by (2) if we plug into the equation the corresponding equilibrium efforts from (6). From Lemma 3, we know that this order of moves arises when the contest organiser chooses to charge the early mover a price $c \in \left[0, \frac{v_1+v_2}{2v_1}\right]$.

In the second possible scenario—when only player 2 moves early and player 1 follows—the contest organiser's expected payoff can be found if we substitute (10) into (4) where $i = 2$. As stated in Lemma 3, this sequence of moves is followed by the two players if the contest organiser chooses a price $c \in \left(\frac{v_1+v_2}{2v_1}, 1\right]$. It is obvious that, when choosing the level of price to set, the contest organiser should take into consideration not only the players' corresponding efforts but also their agreed order of moves induced by this price. Therefore, the contest organiser, in order to set the optimal level of price, should determine her expected payoff in both cases and compare them with each other. The above analysis leads to the following proposition which includes the main result of the paper thus far.

Proposition 1 *In equilibrium, the contest organiser sets for the player that exerts effort early a price* $c \in \left[0, \frac{v_1+v_2}{2v_1}\right]$. *Both players agree to move early and the contest organiser's expected payoff is* $\frac{v_1 v_2}{v_1+v_2}$.

As we have already seen above, if both players move early at the second stage, their efforts depend on the level of the price they pay. However, because both of them pay the same price, the simultaneous game generated is symmetric in the effort costs and asymmetric only in the players' prize valuations. This implies that the two players' expected payoffs are independent of the level of price. Thus, the contest organiser's only goal is to set a price that solely affects the players' sequence of moves. It follows that as the contest organiser has a greater expected payoff—equal to the one in a "typical" asymmetric Tullock contest—when both players move early rather than sequentially, she chooses any level of price that incentivises the two players to follow this particular order of moves.

The result from Proposition 1 is also depicted in Fig. 1 where we illustrate the contest organiser's expected payoff as a function of the price. Beginning from $c = 0$ and simultaneous move by the two players, we observe that the expected payoff is constant in c and at its maximum level up to the point where the players decide to move sequentially and the contest designer's payoff begins to decrease.

4 Simultaneous game with favouritism

Thus far, we have made the assumption that when the two players decide to move early, they both pay the same price. This assumption makes the contest organiser and players' payoffs in the simultaneous game independent of the level of price.

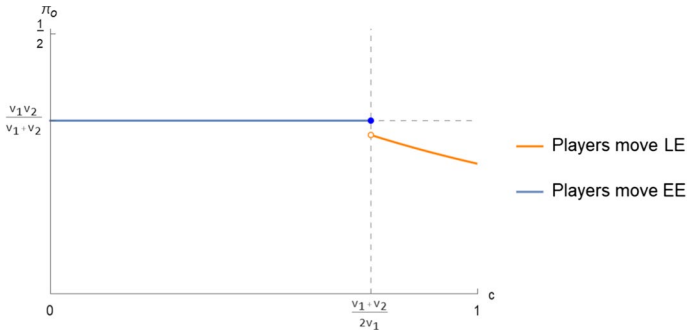

Fig. 1 Contest organiser's payoff as a function of the price

Hence, the contest organiser is indifferent among the levels of the price she may set as long as they induce both players to make effort early in the contest.

In this section, we modify the benchmark model and assume that even when both players move early, still only one of them pays the reduced price. In particular, let us assume that when the two players decide to exert efforts early, player 1 receives the low price with a probability $p \in [0, 1]$ while player 2 receives it with a probability $1 - p$.[13] As a consequence, player 1's expected payoff in the simultaneous game is given by

$$p\left(\frac{x_1}{x_1 + x_2}v_1 - cx_1\right) + (1 - p)\left(\frac{x_1}{x_1 + x_2}v_1 - x_1\right).$$

The first term of the sum above is the probability that player 1 receives the reduced price multiplied by her payoff if she receives it; the second term is the probability that player 2 receives the reduced price times player 1's payoff if she pays a price equal to 1. Simplifying, we can express player 1's expected utility as

$$\tilde{\pi}_{1,E} = \frac{x_1}{x_1 + x_2}v_1 - pcx_1. \tag{11}$$

Similarly, player 2's expected payoff in the simultaneous game can be expressed as[14]

$$\tilde{\pi}_{2,E} = \frac{x_2}{x_1 + x_2}v_2 - (1 - p)cx_2. \tag{12}$$

The rest of the analysis is similar to the one in the previous section.[15] However, now, in order to determine the equilibrium of the timing game and the optimal price

[13] As already discussed in the Introduction, this probability can also be viewed as the contest organiser's favouritism towards a specific player. For instance, in the lobbying case, if $p > \frac{1}{2}$, the politician— possibly because of aligned political preferences—treats group 1 more favourably than group 2.

[14] Observing the two payoff functions, we can notice that if we set $pc = \kappa$, the game becomes similar to a game where the contest organiser sets a different price for each player, i.e., price κ for player 1 and price $c - \kappa$ for player 2.

[15] See Appendix 2 for the complete analysis of the game.

chosen by the contest organiser, we have to use the equilibrium expected payoffs of the new simultaneous game, namely,

$$\tilde{\pi}^*_{1,E} = \frac{\left[p + (1-p)c\right]^2 v_1^3}{\left\{\left[p + (1-p)c\right]v_1 + \left[1 - (1-c)p\right]v_2\right\}^2}$$

and

$$\tilde{\pi}^*_{2,E} = \frac{\left[1 - (1-c)p\right]^2 v_2^3}{\left\{\left[p + (1-p)c\right]v_1 + \left[1 - (1-c)p\right]v_2\right\}^2}.$$

The following proposition presents the solution to the timing game.

Proposition 2 *Let* $r = \frac{1}{4}\sqrt{\frac{9p^2v_1^2+p^2v_2^2-6p^2v_1v_2-2pv_1^2+6pv_1v_2+v_1^2}{p^2v_1^2}} - \frac{(1-p)v_1-pv_2}{4pv_1} > 0$. *In the equilibrium of the timing game, if* $c \in [0, r]$, *both players move early* (E, E). *If* $c \in (r, 1]$, *player* 2 *moves early and player* 1 *moves late* (L, E).

We observe that, differently from the benchmark model, the price threshold below which the two players decide to play simultaneously does not depend only on their prize valuations but also on their probability of receiving the low price. Further investigation of the price threshold gives us the following result.

Corollary 4.1 *The price threshold r is decreasing in* v_1 *and increasing in* v_2, p.

Corollary 4.1 states that the price threshold decreases as player 1's prize valuation goes up. Given that player 2 never prefers to move late, this result indicates that the stronger player 1 becomes, the lower the level of the price she requires in order to agree to move early together with player 2. On the other hand, as player 2 becomes stronger, player 1 feels more threatened and, therefore, is willing to move early even if the level of the price offered is greater. Lastly, the price threshold below which the two players decide to move early increases as the probability that the low price will be paid by player 1 goes up. This result is intuitive as the greater the probability that player 1 pays the low price in the simultaneous game, the greater the level of price that induces her to exert effort early together with her opponent.

Regarding the contest organiser's strategy, in contrast to the case of Sect. 4, now, we can determine the exact optimal level of the price set by the contest organiser. This occurs because, in this case, the price affects the contest organiser's payoff also when the players move simultaneously. Thus, it is not sufficient for the contest organiser only to induce the players to move simultaneously but, also, she has to determine the payoff maximising level of price when the players decide to move simultaneously. Particularly, while—similarly to the benchmark model—the contest organiser's maximisation problem is given by (4) when the players move sequentially, in the case of an early move by the two players, the contest organiser's expected payoff is now given by

$$\tilde{\pi}_{o,E} = p\left(c\tilde{x}^*_{1,E} + \tilde{x}^*_{2,E}\right) + (1-p)\left(\tilde{x}^*_{1,E} + c\tilde{x}^*_{2,E}\right), \tag{13}$$

where, from Appendix 2,

$$\tilde{x}^*_{1,E} = \frac{\left[p + (1-c)p\right]v_1^2 v_2}{\left\{\left[p + (1-p)c\right]v_1 + \left[1 - (1-c)p\right]v_2\right\}^2}$$

and

$$\tilde{x}^*_{2,E} = \frac{\left[1 - (1-c)p\right]v_1 v_2^2}{\left\{\left[p + (1-p)c\right]v_1 + \left[1 - (1-c)p\right]v_2\right\}^2}.$$

The following result emerges.

Proposition 3 *In equilibrium, the contest organiser offers to the player that exerts effort early a price*

$$\tilde{c} = \begin{cases} \frac{(1-p)v_2 - pv_1}{(1-p)v_1 - pv_2}, & \text{if } p < \frac{v_2}{v_1+v_2} \\ 0, & \text{if } \frac{v_2}{v_1+v_2} \leq p < \frac{1}{2} \\ \tilde{c}' \in [0, r], & \text{if } p = \frac{1}{2} \\ r, & \text{if } p > \frac{1}{2} \end{cases}$$

The result in Proposition 3 which is illustrated in Fig. 2, indicates that when in the simultaneous game player 2 receives the low price with a probability sufficiently greater than player 1's, the contest organiser's optimal strategy is to set an intermediate level of price (Fig. 2a). This level depends on the probability that each player receives the low price as well as on the difference between the two players' prize valuations. As p approaches $\frac{v_2}{v_1+v_2}$, the optimal price reaches its minimum possible level and remains constant up to the point where the two players' probabilities of receiving the low price become almost equal (Fig. 2b). On the other hand, if player 1 receives the low price with a probability greater than player 2's, the contest organiser may want to favour the early mover the least possible. Thus, she sets the maximum level of price that induces an early move by both players (Fig. 2c). Notice that if the two players have the same probability of receiving the low price, then, $r = \frac{v_1+v_2}{2v_1}$ and the case coincides with the one of the benchmark model in Sect. 3.

Corollary 4.2 *If $p < \frac{v_2}{v_1+v_2}$, $\tilde{c}$ is decreasing in v_1, p and increasing in v_2.*

In Appendix 1, we show that for $p < \frac{v_2}{v_1+v_2}$, as p increases, then, $\tilde{c}$ goes down. This means that for low levels of p, the greater the probability that player 1 receives the low price, the lower the level of the price set by the contest organiser. Moreover,

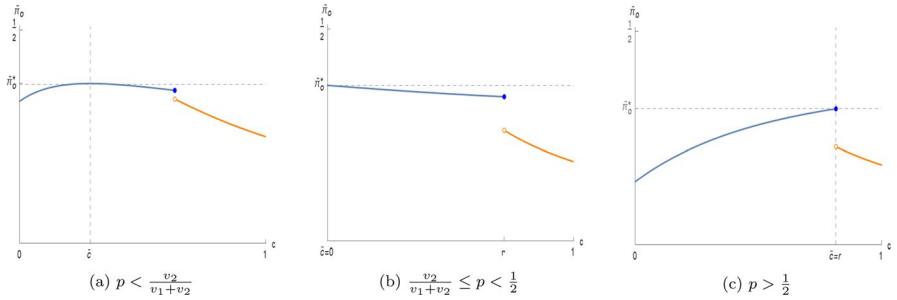

Fig. 2 Contest organiser's expected payoff with favouritism ($v_1 = 1, v_2 = \frac{1}{2}$, orange line: *LE*, blue line: *EE*)

it is not difficult to verify that $\tilde{c}$ is decreasing in v_1 and increasing in v_2. As player 1 becomes stronger, the contest organiser finds it optimal to favour player 2 and given that the latter receives the advantage with a high probability, the contest organiser increases the level of this advantage. On the other hand, as player 2 becomes stronger—and given that she receives the low price with a sufficiently high probability—the contest organiser finds it optimal to moderate player 2's potential *ex post* advantage by increasing the offered price.

Observing Fig. 2, we suspect that the contest organiser's expected payoff declines as p increases. To verify this, in Fig. 3, we illustrate how the contest organiser's equilibrium expected payoff, $\tilde{\pi}_o^*$, behaves as p varies. We observe that when $p < \frac{v_2}{v_1+v_2}$ and the optimal price is $\frac{(1-p)v_2-pv_1}{(1-p)v_1-pv_2}$, the contest organiser's equilibrium expected payoff is constant and equal to $\frac{v_1+v_2}{4}$. As p increases further and the optimal price becomes equal to 0, the equilibrium expected payoff decreases at an increasing rate up to the point where $p = \frac{1}{2}$ and the payoff becomes equal to $\frac{v_1 v_2}{v_1+v_2}$, i.e., the one of the benchmark model (or, equivalently, the one of the simultaneous Tullock contest with no different prices). Finally, as p exceeds $\frac{1}{2}$ and the optimal price is equal to r, the equilibrium payoff continues to decrease but at a decreasing rate. It follows that the greater the contest organiser's favouritism towards the weak player, the greater the expected payoff generated by her optimal choice of price.

5 Conclusion

We have considered a two-player contest in which the players have the opportunity to choose whether to exert effort early or late. In the absence of any intervention by the contest organiser, it is already known from the related literature that the players choose to move sequentially. Particularly, they agree in a sequence of moves where the weak player makes effort early and the strong player replies after having observed the weak player's chosen level of effort. This order of moves leads to the lowest possible total effort and is, therefore, beneficial for

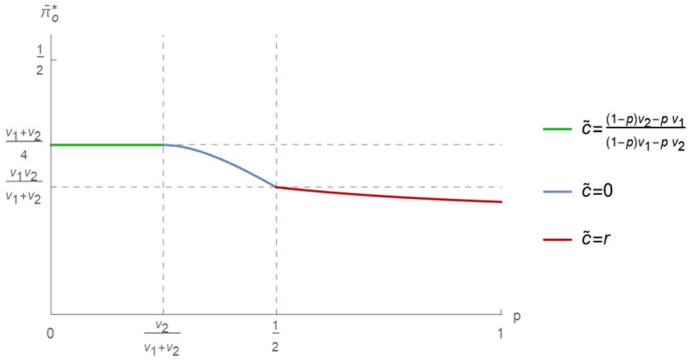

Fig. 3 Contest organiser's expected payoff as a function of p at the optimal levels of price ($v_1 = 1, v_2 = \frac{1}{3}$)

both players. However, an organiser of such a contest prefers a different order of moves which may generate a greater total effort and, as a consequence, a greater payoff for her.

In this study, we have made the contest organiser an active part of the game by allowing her to offer a lower price per unit of effort to the player that commits to make effort early. By doing so, she is able to increase her expected payoff by inducing the two players to agree to move in a different order. Specifically, while the weak player has always the incentive to move early and pay the low price, a sufficiently low level of price is attractive to player 1 inducing her to also move early together with her opponent.

We have also analysed the case where if the players make effort without knowing their opponents' efforts (simultaneous game), only one of them receives the low price. A new strength asymmetry between the players—apart from the one in the prize valuations— is then the difference in the contest organiser's favouritism towards the two players. This asymmetry might lead to a situation where the player who values the prize lower might have an expected payoff greater than her opponent's if the contest organiser's favouritism towards the latter is sufficiently low. Regarding the contest organiser, when choosing the level of price to offer to the early mover, she also takes into consideration this bias. Investigating her optimal strategy, we have established that she prefers a situation where the weak player receives the low price with a greater probability than the strong one when the two players move early. The intuition is that the greater the favouritism towards the weak player, the more "symmetric" the contest becomes which makes the contest organiser better off.

A natural extension of this study would be the investigation of the case with more than two players. It would also be interesting to endogenise the probability that the players receive the low price in the simultaneous game so that it is under the contest organiser's control. We leave the exploration of these possible extensions for future research.

Appendix 1: Proofs

Proof of Lemma 1 First, we maximise player 2's payoff function. Player 2 chooses her effort taking into account player 1's effort who moved earlier. The FOC of (9) is

$$\frac{x_{1,12}}{\left(x_{1,12}+x_{2,12}\right)^2}v_2 - 1 = 0$$

which yields the reaction function $x_{2,12}\left(x_{1,12}\right) = \sqrt{x_{1,12}v_2} - x_{1,12}$. Following the analysis in Leininger (1993), since it must be $x_{2,12} \geq 0$, i.e., the effort cannot be negative, player 2 makes effort

$$x_{2,12}\left(x_{1,12}\right) = \begin{cases} \sqrt{x_{1,12}v_2} - x_{1,12}, & \text{if } x_{1,12} \leq v_2 \\ 0, & \text{if } x_{1,12} \geq v_2. \end{cases}$$

Player 1 chooses her level of effort to exert anticipating player 2's optimal response. Substituting player 2's reaction function into (8), we can express player 1's payoff function as

$$\pi_{1,12}\left(x_{1,12}\right) = \begin{cases} \frac{\sqrt{x_{1,12}}}{\sqrt{v_2}}v_1 - cx_{1,12}, & \text{if } x_{1,12} \leq v_2 \\ v_1 - cx_{1,12}, & \text{if } x_{1,12} \geq v_2. \end{cases}$$

For $x_{1,12} \leq v_2$, differentiating the above function with respect to $x_{1,12}$ and setting the derivative equal to zero, we obtain the FOC of player 1's maximisation problem, namely,

$$\frac{v_1}{2\sqrt{v_2}\sqrt{x_{1,12}}} - c = 0.$$

Solving for $x_{1,12}$, we find that player 1's optimal level of effort is $x^*_{1,12} = \frac{v_1^2}{4c^2v_2}$. Substituting $x^*_{1,12}$ into player 2's reaction function, we obtain that she replies with an effort $x^*_{2,12} = \frac{v_1(2cv_2-v_1)}{4c^2v_2}$. For $x_{1,12} \geq v_2$, it is not difficult to observe that player 1's payoff function is decreasing in $x_{1,12}$ and, therefore, her optimal level of effort is the minimum possible, i.e., equal to v_2. Finally, we have that $x^*_{1,12} \leq v_2 \implies \frac{v_1^2}{4c^2v_2} \leq v_2 \implies c \geq \frac{v_1}{2v_2}$ implying that $x^*_{1,12} \geq v_2$ if $c \leq \frac{v_1}{2v_2}$, where $v_1 < 2v_2$. For $v_1 \geq 2v_2$, we have that $x^*_{1,12} \leq v_2$ for all c. Substituting the equilibrium efforts for each case, we obtain the equilibrium payoffs stated in the proposition. The proof of Lemma 2 is analogous and, thus, omitted. □

Proof of Lemma 3 To prove the proposition, we study each possible case of Table 1 individually according to Lemmas 1 and 2.[16] Note that $\pi_{i,sim}$ remains the same in each case and only the payoffs of the sequential games may vary.

[16] Keep in mind that all the payoffs included in the proof are the equilibrium expected payoffs. The superscript asterisks are omitted for simplification of notation.

Let us first assume that $v_1 \geq 2v_2$. If player 1 moves early and player 2 follows, player 1 and 2's expected payoffs are equal to $\pi_{1,12} = v_1 - cv_2$ and $\pi_{2,12} = 0$, respectively. Nevertheless, if player 2 moves early and player 1 follows, the pair of the two players' expected payoffs vary depending on the interval c lies in. Particularly, if $c \leq \frac{v_2}{2v_1}$, then, $\pi_{1,21} = 0$ and $\pi_{2,21} = v_2 - cv_1$ while if $c > \frac{v_2}{2v_1}$, then, $\pi'_{1,21} = \frac{(v_2-2cv_1)^2}{4c^2v_1}$ and $\pi'_{2,21} = \frac{v_2^2}{4cv_1}$. We study the two cases separately.

- *Case* $c \in \left[0, \frac{v_2}{2v_1}\right]$: Let player 1 move early. As $\pi_{2,sim} > \pi_{2,12} = 0$, player 2's response is to move early as well. If player 1 moves late, we have that $\pi_{2,21} > \pi_{2,sim}$ and, therefore, player 2 moves early. Thus, player 2 has the dominant strategy E. Regarding player 1, if player 2 moves early, then, $\pi_{1,sim} > \pi_{1,21} = 0$ and, therefore, player 1 moves early, too. If player 2 moves late, $\pi_{1,12} > \pi_{1,sim}$ and, thus, player 1 moves early as well, meaning that player 1 has also the dominant strategy E. The equilibrium of the game is (E, E).
- *Case* $c \in \left(\frac{v_2}{2v_1}, 1\right]$: It is not difficult to verify that player 2 has a dominant strategy to move early in this case, too. Nevertheless, player 1 does not have a dominant strategy. Actually, when player 2 moves early, player 1 moves late only if $c > \frac{v_1+v_2}{2v_1}$ and early otherwise. Consequently, the equilibrium of the game is (E, E) if $c \leq \frac{v_1+v_2}{2v_1}$ and (L, E) if $c > \frac{v_1+v_2}{2v_1}$.

Let us now assume that $v_1 < 2v_2$. We have three cases with different pairs of payoffs in the sequential game: $c \in \left[0, \frac{v_2}{2v_1}\right]$, $c \in \left[\frac{v_2}{2v_1}, \frac{v_1}{2v_2}\right)$ and $c \in \left[\frac{v_1}{2v_2}, 1\right]$. We can observe that the first two cases lead to games similar to the ones when $v_1 \geq 2v_2$. Now, for the case where $c \in \left[\frac{v_1}{2v_2}, 1\right]$, in contrast to the previous ones, the payoffs when player 1 moves early and player 2 moves late change. Actually, we have that $\pi'_{1,12} = \frac{v_1^2}{4cv_2}$ and $\pi'_{2,12} = \frac{(v_1-2cv_2)^2}{4c^2v_2}$. Again, in this case, player 2's dominant strategy is to move early. However, $\pi'_{1,21} > \pi_{1,sim}$, i.e., player 1 responds with a late move only if $v_2 \leq \frac{\sqrt{5}-1}{2}v_1$ or if $v_2 > \frac{\sqrt{5}-1}{2}v_1$ and $c > \frac{v_1+v_2}{2v_1}$. On the other hand, if $v_2 > \frac{\sqrt{5}-1}{2}v_1$ and $c \leq \frac{v_1+v_2}{2v_1}$, player 1 chooses to move early.

Proceeding to investigate the equilibrium of the timing game, we compare with each other the players' expected payoffs in all the above possible scenarios. For $v_2 < \frac{\sqrt{5}-1}{2}v_1$, we have that $\frac{v_1}{2v_2} > \frac{v_1+v_2}{2v_1}$. In this case, if $c \leq \frac{v_1+v_2}{2v_1}$, the equilibrium is (E, E) while if $c > \frac{v_1+v_2}{2v_1}$, the equilibrium is (L, E). For $v_2 \geq \frac{\sqrt{5}-1}{2}v_1$, we have that $\frac{v_1}{2v_2} \leq \frac{v_1+v_2}{2v_1}$, where the equilibrium behaves similarly to the previous case and the proof is complete. □

Proof of Proposition 1 Firstly, let us consider the case where the contest organiser sets a price $c \in \left(\frac{v_1+v_2}{2v_1}, 1\right]$. She anticipates, then, that at the second stage, the two players will move sequentially, i.e., player 2 will move early and player 1 will move late. Her expected payoff is, therefore, given by (4). Substituting the players' second-stage equilibrium efforts into (4), we have

$$\pi_{o,21}\left(x^*_{1,21}, x^*_{2,21}\right) = \frac{v_2\left[2c\delta v_1 + v_2(c-\delta)\right]}{4c^2 v_1}.$$

Differentiating the above expression with respect to c, we obtain

$$\frac{\partial \pi_{o,21}\left(x^*_{1,21}, x^*_{2,21}\right)}{\partial c} = \frac{v_2\left[v_2(2\delta - c) - 2\delta c v_1\right]}{4c^3 v_1}.$$

The above expression is negative if $c > \frac{2\delta v_2}{2\delta v_1 + v_2}$. Moreover, we have that $\frac{v_1+v_2}{2v_1} - \frac{2\delta v_2}{2\delta v_1+v_2} = \frac{2\delta v_1^2+(1-2\delta)v_1v_2+v_2^2}{2v_1(2\delta v_1+v_2)} > \frac{2\delta v_1(v_1-v_2)}{2v_1(2\delta v_1+v_2)} \geq 0$ which is true because of the assumptions that $\delta \in [0,1]$ and $v_1 > v_2$. We conclude that if $c > \frac{v_1+v_2}{2v_1}$, then, $\frac{\partial \pi_{o,21}}{\partial c} < 0$. This implies that for $c \in \left(\frac{v_1+v_2}{2v_1}, 1\right]$, the contest organiser's payoff function is decreasing in c and, therefore, maximised when $c \to \frac{v_1+v_2}{2v_1}$, yielding for the contest organiser expected payoff $\lim_{c\to\frac{v_1+v_2}{2v_1}} \pi_{o,21}\left(x^*_{1,21}, x^*_{2,21}\right) = \frac{v_2\left(2\delta v_1^2+v_1v_2+v_2^2\right)}{2\left(v_1+v_2\right)^2}$.

Now, assume that the contest organiser sets a price $c \in \left[0, \frac{v_1+v_2}{2v_1}\right]$. If she does so, both players will choose to move early at the next stage. Her expected payoff in this case is given by substituting (6) into (2), namely,

$$\pi_{o,E}\left(x^*_{1,E}, x^*_{2,E}\right) = \frac{v_1 v_2}{v_1 + v_2}.$$

We observe that the contest organiser's expected payoff is independent of price. Thus, she is indifferent among all the levels of price included in $\left[0, \frac{v_1+v_2}{2v_1}\right]$ as for any c in this interval, the players will exert the same levels of effort and the contest organiser will gain the same payoff.

Comparing the contest organiser's payoffs in the two cases mentioned above, we have $\frac{v_2\left(2\delta v_1^2+v_1v_2+v_2^2\right)}{2\left(v_1+v_2\right)^2} - \frac{v_1v_2}{v_1+v_2} = \frac{v_2\left[v_2^2-2(1-\delta)v_1^2-v_1v_2\right]}{2\left(v_1+v_2\right)^2} < \frac{v_2^2\left(v_2-v_1\right)}{2\left(v_1+v_2\right)^2} < 0$ which is true by assumption. We conclude that the contest organiser is better off when the two players move early at the second stage. In order to achieve this particular order of moves, she sets for the early mover a price $c \in \left[0, \frac{v_1+v_2}{2v_1}\right]$. □

Proof of Proposition 2 The proof of the proposition is similar to the one of Lemma 3. However, here, for the cases (E, E) and (L, L), instead of the players' payoffs from the

typical Tullock contest considered in the benchmark model, we consider the ones of the new game, namely, $\tilde{\pi}_{i,sim}$, $i = 1, 2$.

Following the proof of Lemma 3, assume first that $v_1 \geq 2v_2$. If $c \leq \frac{v_2}{2v_1}$, similarly to the benchmark model, we have that $\pi_{1,12} > \tilde{\pi}_{1,sim}$ and $\pi_{2,21} > \tilde{\pi}_{2,sim}$. Therefore, both players prefer to move early and the equilibrium is (E, E). If $c > \frac{v_2}{2v_1}$, we have that $\tilde{\pi}_{2,sim} < \pi'_{2,21}$ and $\tilde{\pi}_{2,sim} > 0$, i.e., player 2 still has a dominant strategy to move early. On the other hand, regarding player 1, we have that $\tilde{\pi}_{1,sim} \geq \pi'_{1,21}$ only if $c \leq r$ which implies that in this case, she prefers to move early and the equilibrium of the game is (E, E) while if $c > r$, she prefers to move late and the equilibrium of the game is (L, E).

Assume now that $v_1 < 2v_2$. We consider the same three cases as in the proof of Lemma 3, i.e., $c \in \left[0, \frac{v_2}{2v_1}\right]$, $c \in \left[\frac{v_2}{2v_1}, \frac{v_1}{2v_2}\right)$ and $c \in \left[\frac{v_1}{2v_2}, 1\right]$. It is not difficult to check that in all these cases, player 2's dominant strategy is to move early. We study, then, only player 1's strategy separately for each case.

- *Case* $c \in \left[0, \frac{v_2}{2v_1}\right]$: This case coincides with the respective one from the proof of Lemma 3 and, thus, the equilibrium is (E, E).
- *Case* $c \in \left[\frac{v_2}{2v_1}, \frac{v_1}{2v_2}\right)$: Given that player 2 always moves early, the only possible payoffs for player 1 are $\pi'_{1,21}$ if she moves late and $\tilde{\pi}_{1,sim}$ if she moves early together with player 2. Comparing these two payoffs with each other, we find that for $v_2 \leq \frac{\sqrt{3}v_1}{3}$, if $c \in \left[\frac{v_2}{2v_1}, r\right)$, player 1 prefers to move early and the equilibrium is (E, E) while if $c \in \left[r, \frac{v_1}{2v_2}\right)$, she prefers to move late and the equilibrium is (L, E). For $\frac{\sqrt{3}v_1}{3} < v_2 < \frac{\sqrt{2}v_1}{2}$, let $\bar{p} = \frac{v_2(v_1^2 - 2v_2^2)}{(2v_2 - v_1)(v_1^2 + v_1 v_2 - v_2^2)}$. If $c \in \left[r, \frac{v_1}{2v_2}\right)$ and $p < \bar{p}$, the equilibrium is (L, E) while if $p \geq \bar{p}$ or if $p < \bar{p}$ and $c \in \left[\frac{v_2}{2v_1}, r\right)$ the equilibrium is (E, E). Lastly, for $v_2 \geq \frac{\sqrt{2}v_1}{2}$, player 1 always prefers to move early and the equilibrium is (E, E).
- *Case* $c \in \left[\frac{v_1}{2v_2}, 1\right]$: In this case, for $v_2 \leq \frac{\sqrt{3}v_1}{3}$ player 1 moves late and the equilibrium is (L, E). Now, for $\frac{\sqrt{3}v_1}{3} < v_2 \leq \frac{\sqrt{2}v_1}{2}$, if $p \leq \bar{p}$ or if $p > \bar{p}$ and $c \in \left[\frac{v_1}{2v_2}, r\right)$, the equilibrium is (L, E) while if $p > \bar{p}$ and $c \in \left[r, 1 - \frac{v_1}{2v_2}\right)$, the equilibrium is (E, E). Finally, for $v_2 > \frac{\sqrt{2}v_1}{2}$, if $c \in \left[\frac{v_1}{2v_2}, r\right]$, the equilibrium is (E, E) while if $c \in (r, 1]$, the equilibrium is (L, E).

From the above analysis, combining all the possible cases, we can conclude that the equilibrium of the timing game is (E, E) if $c \in [0, r]$ and (L, E) if $c \in (r, 1]$. □

Proof of Corollary 4.1 Differentiating r with respect to v_1, we obtain

$$\frac{\partial r}{\partial v_1} = -\frac{\left[\left(\sqrt{b}-3\right)p+3\right]v_1v_2 + pv_2^2}{4\sqrt{b}pv_1^3}$$

where $b = \frac{(9p^2-2p+1)v_1^2+[6(1-p)v_1+pv_2]pv_2}{p^2v_1^2} > 0$. To show that the above derivative is negative, it is sufficient to show that $\left(\sqrt{b}-3\right)p+3>0 \implies \sqrt{b} > -\frac{3(1-p)}{p}$, which is true. Thus, $\frac{\partial r}{\partial v_1} < 0$.

Differentiating r with respect to v_2, we have

$$\frac{\partial r}{\partial v_2} = \frac{\left[\left(\sqrt{b}-3\right)p+3\right]v_1 + pv_2}{4\sqrt{b}pv_1^2}$$

where it follows from previously that $\frac{\partial r}{\partial v_2} > 0$.

Finally, differentiating r with respect to p, we get

$$\frac{\partial r}{\partial p} = -\frac{\left[1-\left(1+\sqrt{b}\right)p\right]v_1 + 3pv_2}{4p^3\sqrt{b}v_1}.$$

Let us assume that $\frac{\partial r}{\partial p} \le 0$. We have, then, that

$$\begin{aligned}
&\frac{\left[1-\left(1+\sqrt{b}\right)p\right]v_1 + 3pv_2}{4p^3\sqrt{b}v_1} \ge 0 \implies \sqrt{b} \le \frac{(1-p)v_1+3pv_2}{pv_1} \\
&\implies \frac{v_1^2 - 2pv_1^2 + 9p^2v_1^2 + 6pv_1v_2 - 6p^2v_1v_2 + p^2v_2^2}{p^2v_1^2} \le \frac{\left[(1-p)v_1+3pv_2\right]^2}{p^2v_1^2} \\
&\implies \quad 8p^2v_1^2 - 8p^2v_2^2 \le 0 \implies \left(v_1-v_2\right)\left(v_1+v_2\right) \le 0
\end{aligned}$$

which is not true, given the initial assumption that $v_1 > v_2$. Thus, $\frac{\partial r}{\partial p} > 0$. □

Proof of Proposition 3 We know from Proposition 2 that, at the second stage, the order of moves decided by the two players is either (E, E) or (L, E). Particularly, the decided order of moves is the former if $c \in [0, r]$ and the latter if $c \in (r, 1]$. Beginning from the second case, the contest organiser's expected payoff is given by (4) which—from the proof of Proposition 1—is a decreasing function in $c \in \left(\frac{2\delta v_2}{2\delta v_1+v_2}, 1\right]$. It can also be shown that

$$r \equiv \frac{1}{4}\sqrt{\frac{9p^2v_1^2 + p^2v_2^2 - 6p^2v_1v_2 - 2pv_1^2 + 6pv_1v_2 + v_1^2}{p^2v_1^2}} - \frac{(1-p)v_1 - pv_2}{4pv_1} > \frac{2\delta v_2}{2\delta v_1 + v_2}$$

meaning that the expected payoff is a decreasing function also in $(r, 1]$. Thus, in this case, the contest organiser sets a price $c \to r$.

In the case where $c \leq r$, both players move early and, therefore, the contest organiser maximises (13) with respect to c. The FOC of the maximisation problem is

$$\frac{\partial \tilde{\pi}_{o,E}}{\partial c} = \frac{(1-2p)v_1 v_2 \left(v_1 + v_2\right)\left\{v_2\left[1-(1-c)p\right] - v_1\left[c(1-p)+p\right]\right\}}{\left\{v_1\left[(1-p)c+p\right] + v_2\left[1-(1-c)p\right]\right\}^3} = 0$$

where the denominator is a positive quantity, i.e., $v_1\left[(1-p)c+p\right] + v_2\left[1-(1-c)p\right] > 0$. Solving the FOC for c, we find that $c = \frac{(1-p)v_2 - pv_1}{(1-p)v_1 - pv_2}$, where $p < \frac{v_2}{v_1+v_2}$ since it must be $0 < c \leq r$. Now, for $p \geq \frac{v_2}{v_1+v_2}$, we have that $v_2\left[1-(1-c)p\right] - v_1\left[c(1-p)+p\right] < 0 \implies p > \frac{v_2 - cv_1}{(1-c)\left(v_1+v_2\right)}$. Moreover, $\frac{v_2}{v_1+v_2} \geq \frac{v_2-cv_1}{(1-c)\left(v_1+v_2\right)} \implies (1-c)v_2\left(v_1+v_2\right) \geq \left(v_1+v_2\right)\left(v_2-cv_1\right) \implies c\left(v_1^2 - v_2^2\right) \geq 0$, which is true by assumption. Therefore, $p \geq \frac{v_2}{v_1+v_2} \implies p \geq \frac{v_2-cv_1}{(1-c)\left(v_1+v_2\right)}$ which, in turn, implies that $v_2\left[1-(1-c)p\right] - v_1\left[c(1-p)+p\right] < 0$. Hence, the sign of $\frac{\partial \tilde{\pi}_{o,E}}{\partial c}$ depends on the sign of $1-2p$. In particular,

$$\frac{\partial \tilde{\pi}_{o,E}}{\partial c} \begin{cases} < 0, & \text{if } \frac{v_2}{v_1+v_2} \leq p < \frac{1}{2} \\ > 0, & \text{if } p > \frac{1}{2} \end{cases}$$

which indicates that $\tilde{\pi}_{o,E}$ is decreasing in c if $\frac{v_2}{v_1+v_2} \leq p < \frac{1}{2}$ and increasing in c if $p > \frac{1}{2}$. Thus, in the first case, the contest organiser sets a price $\tilde{c} = 0$ while in the second case, she sets a price $\tilde{c} = r$. For the case where $p = \frac{1}{2}$, the contest organiser's expected payoff is simplified to $\frac{v_1 v_2}{v_1+v_2}$ which is independent of c. Thus, it is sufficient that the price set by the contest organiser induces simultaneous moves by the players, i.e., $\tilde{c} \in [0, r]$. □

Proof of Corollary 4.2 For $p < \frac{v_2}{v_1+v_2}$, we have

$$\frac{\partial \tilde{c}}{\partial p} = \frac{v_2^2 - v_1^2}{\left[(1-p)v_2 - pv_1\right]^2} < 0, \quad \frac{\partial \tilde{c}}{\partial v_1} = -\frac{(1-2p)v_2}{\left[(1-p)v_1 - pv_2\right]^2} < 0$$

and

$$\frac{\partial \tilde{c}}{\partial v_2} = \frac{(1-2p)v_1}{\left[(1-p)v_1 - pv_2\right]^2} > 0.$$

□

Appendix 2: Analysis of the simultaneous contests with favouritism

For player 1, the FOC of (11) is

$$p(1-c)+\frac{x_2}{\left(x_1+x_2\right)^2}v_1-1=0$$

which gives the reaction function $x_1\left(x_2\right)=\sqrt{\frac{x_2v_1}{1-(1-c)p}}-x_2$.

Regarding player 2, the FOC of (12) is

$$pc+\frac{x_1}{\left(x_1+x_2\right)^2}v_2-c-p=0$$

which gives the reaction function $x_2\left(x_1\right)=\sqrt{\frac{x_1v_2}{p+(1-p)c}}-x_1$.

Substituting the second reaction function into the first one, we have

$$\begin{aligned}
x_1 &= \sqrt{\frac{\left(\sqrt{\frac{x_1v_2}{p+(1-p)c}}-x_1\right)v_1}{1-(1-c)p}}-\sqrt{\frac{x_1v_2}{p+(1-p)c}}+x_1\\
&\Longrightarrow \frac{\left(\sqrt{\frac{x_1v_2}{p+(1-p)c}}-x_1\right)v_1}{1-(1-c)p}=\frac{x_1v_2}{p+(1-p)c}\\
&\Longrightarrow x_1v_2\left[1-(1-c)p\right]=\left[p+(1-p)c\right]v_1\left(\sqrt{\frac{x_1v_2}{p+(1-p)c}}-x_1\right)\\
&\Longrightarrow x_1^2\left\{\left[p+(1-p)c\right]v_1+\left[1-(1-c)p\right]v_2\right\}^2=x_1\left[p+(1-c)p\right]v_1^2v_2\\
&\Longrightarrow \tilde{x}_{1,E}^*=\frac{\left[p+(1-c)p\right]v_1^2v_2}{\left\{\left[p+(1-p)c\right]v_1+\left[1-(1-c)p\right]v_2\right\}^2}.
\end{aligned}$$

Substituting the above quantity into player 2's reaction function, we get

$$\tilde{x}_{2,E}^*=\frac{\left[1-(1-c)p\right]v_1v_2^2}{\left\{\left[p+(1-p)c\right]v_1+\left[1-(1-c)p\right]v_2\right\}^2}.$$

Plugging the above equilibrium efforts into (11) and (12), we obtain player 1 and 2's equilibrium expected payoffs, i.e., $\tilde{\pi}_{1,E}^*$ and $\tilde{\pi}_{2,E}^*$, respectively.

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Declarations

Conflict of interest The author declares no conflict of interest for this article.

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